# Depreciation Financing Prudence and the Cross Section of Stock Returns

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October 20, 2025

**Abstract**

This paper studies the asset pricing implications of Depreciation Financing Prudence (DFP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on DFP achieves an annualized gross (net) Sharpe ratio of 0.56 (0.44), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 23 (18) bps/month with a t-statistic of 3.27 (2.67), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth) is 18 bps/month with a t-statistic of 2.66.

# 1 Introduction

Financial markets efficiently incorporate information about firms' production decisions into asset prices, yet the precise mechanisms through which operational choices affect expected returns remain incompletely understood. While extensive literature documents the pricing of investment, profitability, and financial constraints in the cross-section of stock returns, researchers have devoted less attention to how firms' depreciation financing decisions—the extent to which depreciation expenses cover capital expenditures—relate to their cost of capital and risk exposures. This gap persists despite depreciation representing a substantial non-cash expense that directly affects firms' internal financing capacity and investment flexibility.

We propose that Depreciation Financing Prudence (DFP) captures firms' exposure to investment frictions and productivity shocks within a production-based asset pricing framework. In models with costly external finance and investment adjustment costs, firms that maintain higher ratios of depreciation to capital expenditures face lower marginal costs of investment Zhang (2005); Liu et al. (2009). These firms can more readily adjust their capital stock in response to productivity shocks without incurring substantial external financing costs or violating financial constraints. The production-based asset pricing theory predicts that firms with greater investment flexibility command lower risk premia because they can better smooth cash flows across states of nature Cochrane (1991); Berk et al. (1999).

The DFP signal directly maps to firms' conditional risk exposures through the investment channel. Firms with high depreciation relative to capital expenditures maintain a buffer of internal funds that reduces their sensitivity to aggregate productivity shocks and credit market conditions Livdan et al. (2009). During economic downturns when external finance becomes costly and productivity declines, these firms can continue investing by utilizing depreciation-generated cash flows, thereby maintaining operational flexibility. Conversely, firms with low DFP ratios must rely more heavily on external markets to fund investment, exposing them to systematic variation in the cost of capital and aggregate investment opportunities.

Cross-sectional variation in DFP should therefore predict returns through its correlation with firms' loadings on production-based risk factors. The investment-based asset pricing framework of Lin and Zhang (2013) demonstrates that firms' investment-to-assets ratios and profitability jointly determine their exposure to the investment factor, which prices the cross-section of expected returns. Our DFP measure captures a complementary dimension of investment behavior—the sustainability of capital formation through internal resources—that affects firms' sensitivities to shocks in aggregate investment conditions and the marginal product of capital. Firms maintaining prudent depreciation financing exhibit lower covariation with these systematic factors, warranting lower expected returns in equilibrium.

We found that a value-weighted long/short trading strategy based on DFP achieved an annualized gross Sharpe ratio of 0.56 and a net Sharpe ratio of 0.44 after accounting for transaction costs. The strategy generated monthly average excess returns of 28 basis points with a t-statistic of 3.99, demonstrating robust predictive power for the cross-section of stock returns. Most notably, the strategy delivered a monthly alpha of 23 basis points relative to the Fama-French six-factor model with a t-statistic of 3.27, indicating that DFP captures risk exposures not explained by conventional factor models.

The economic significance of our results persisted across different portfolio construction methodologies and firm size categories. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the DFP strategy earned average returns of 37 basis points per month with a t-statistic of 4.10. The strategy's alpha relative to the six-factor model for these large-cap stocks equaled 27 basis points per month with a t-statistic of 3.06, demonstrating that the DFP effect was not confined to small, illiquid securities. Factor loadings revealed that the long/short

DFP portfolio exhibited its most significant exposure to the CMA (investment) factor with a coefficient of 0.32 and t-statistic of 6.82, consistent with our production-based interpretation.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DFP strategy maintained an alpha of 18 basis points per month with a t-statistic of 2.66. The strategy's gross Sharpe ratio of 0.56 exceeded 93% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.44 surpassed 97% of existing strategies after transaction costs. These results established DFP as an economically meaningful and statistically robust predictor of cross-sectional returns that expanded the investment opportunity set beyond existing factors.

Our study contributes to the production-based asset pricing literature by identifying depreciation financing prudence as a novel determinant of cross-sectional expected returns. While Wu et al. (2009) and Lin and Zhang (2013) establish the theoretical foundations for investment-based pricing factors, and Novy-Marx (2013) documents the pricing of gross profitability, our work extends this framework by demonstrating how firms' depreciation-to-investment ratios capture their exposure to productivity and adjustment cost shocks. Unlike the asset growth anomaly of Cooper et al. (2008) or the investment factor in Fama and French (2015), which focus on the level of investment activity, our DFP measure captures the sustainability and internal financing capacity of firms' capital formation processes.

Methodologically, we advance the anomaly testing literature by implementing the comprehensive protocol of Novy-Marx and Velikov (2023b), which ensures robust evaluation of new return predictors through extensive controls for data mining concerns and transaction costs. Our analysis demonstrates that DFP not only survives conventional robustness tests but also improves the mean-variance efficient frontier achievable with existing factor models after accounting for implementation costs. The generalized net alpha framework reveals that DFP would have expanded the investment opportunity set for 80% of the factor model specifications examined, establishing its practical relevance for portfolio construction.

Our findings have broader implications for understanding the interaction between corporate financial policies and asset prices within production economies. The strong predictive power of depreciation financing prudence suggests that investors price firms' operational flexibility and internal financing capacity as distinct sources of systematic risk. This evidence supports theoretical models that emphasize investment frictions and adjustment costs as fundamental determinants of risk premia, while highlighting the importance of non-cash accounting flows in capturing firms' true economic exposures to aggregate shocks.

# 2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. The sample spans several decades, allowing us to examine the depreciation financing prudence signal across various market conditions and business cycles.signal construction relies on two key COMPUSTAT variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that increase a firm's long-term obligations. This item reflects the actual cash inflows from debt financing activities as reported in the statement of cash flows. Depreciation and amortization (DP) measures the non-cash charges against earnings that reflect the allocation of tangible and intangible asset costs over their useful lives. This variable captures the systematic expensing of capital investments and represents a key component of accrual-based earnings adjustments.construct the Depreciation Financing Prudence signal by calculating the year-over-year change in long-term debt issuance scaled by lagged depreciation and amortization: (DLTIS(t) - DLTIS(t-1)) / DP(t-1). This ratio captures the relationship between changes in a firm's debt financing activities and its asset base depreciation, providing insight into how aggressively firms are expanding their debt relative to the deterioration of their existing productive assets. We compute this measure using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both DLTIS and DP, and exclude observations where lagged depreciation equals zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

# 3 Signal diagnostics

Figure 1 plots descriptive statistics for the DFP signal. Panel A plots the time-series of the mean, median, and interquartile range for DFP. On average, the cross-sectional mean (median) DFP is -5.01 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DFP data. The signal's interquartile range spans -2.72 to 2.81. Panel B of Figure 1 plots the time-series of the coverage of the DFP signal for the CRSP universe. On average, the DFP signal is available for 6.14% of CRSP names, which on average make up 7.30% of total market capitalization.

# 4 Does DFP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DFP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DFP portfolio and sells the low DFP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DFP strategy earns an average return of 0.28% per month with a t-statistic of 3.99. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 3.27 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 6.82 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 520 stocks and an average market capitalization of at least $1,591 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 4.04. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -8-22bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.41. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DFP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DFP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DFP, as well as average returns and alphas for long/short trading DFP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the

$80^{\text{th}}$ NYSE percentile), the DFP strategy achieves an average return of 37 bps/month with a t-statistic of 4.10. Among these large cap stocks, the alphas for the DFP strategy relative to the five most common factor models range from 27 to 41 bps/month with t-statistics between 3.06 and 4.67.

# 5 How does DFP perform relative to the zoo?

Figure 2 puts the performance of DFP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the DFP strategy falls in the distribution. The DFP strategy's gross (net) Sharpe ratio of 0.56 (0.44) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DFP strategy (red line).[2] Ignoring trading costs, a \$1 invested in the DFP strategy would have yielded \$3.97 which ranks the DFP strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DFP strategy would have yielded \$2.45 which ranks the DFP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

Table 1, and indicates the ranking of the DFP relative to those. Panel A shows that the DFP strategy gross alphas fall between the 66 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DFP strategy has a positive net generalized alpha for five out of the five factor models. In these cases DFP ranks between the 80 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

# 6 Does DFP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DFP with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DFP or at least to weaken the power DFP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{DFP}$ from Fama-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DFP} DFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{DFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DFP. Stocks are finally grouped into five DFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DFP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DFP signal in these Fama-MacBeth regressions exceed 1.79, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on DFP is 0.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the DFP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DFP strategy earns alphas that range from 19-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DFP trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.66.

## 7 Does DFP add relative to the whole zoo?

Finally, we can ask how much adding DFP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DFP signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes DFP grows to $1211.90.

## 8 Conclusion

This study provides robust evidence that Depreciation Financing Prudence (DFP) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DFP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies. The value-weighted long/short

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DFP is available.

strategy's annualized Sharpe ratio of 0.56 (gross) and 0.44 (net) indicates strong economic significance, while the strategy's ability to generate monthly alphas of 23 basis points relative to the six-factor model underscores its distinct information content. Particularly noteworthy is the signal's resilience when tested against twelve factors including six closely related strategies from the factor zoo, where it maintains a statistically significant alpha of 18 basis points per month (t-statistic = 2.66). These findings suggest that DFP captures a unique dimension of firm financial prudence that is not fully explained by existing factors related to financing decisions, asset growth, or operational efficiency. From a practical perspective, the strategy's net returns after accounting for transaction costs remain economically significant, suggesting potential implementability for institutional investors. The results contribute to the growing literature on the importance of accounting-based signals in asset pricing and highlight the value of examining how firms manage the financing of their depreciation expenses as an indicator of managerial quality and future performance. Future research should explore the underlying economic mechanisms driving the DFP premium, including whether it reflects systematic risk compensation or behavioral biases in how investors process depreciation-related information. Additionally, examining the signal's performance across different market regimes, international markets, and various subsamples could provide insights into its robustness and boundary conditions. Key limitations of this study include the reliance on accounting data quality, potential concerns about data mining despite our rigorous testing protocol, and the need for out-of-sample validation as more data becomes available. Furthermore, while we control for numerous related factors, the possibility remains that DFP proxies for an unidentified risk factor that future research may uncover.

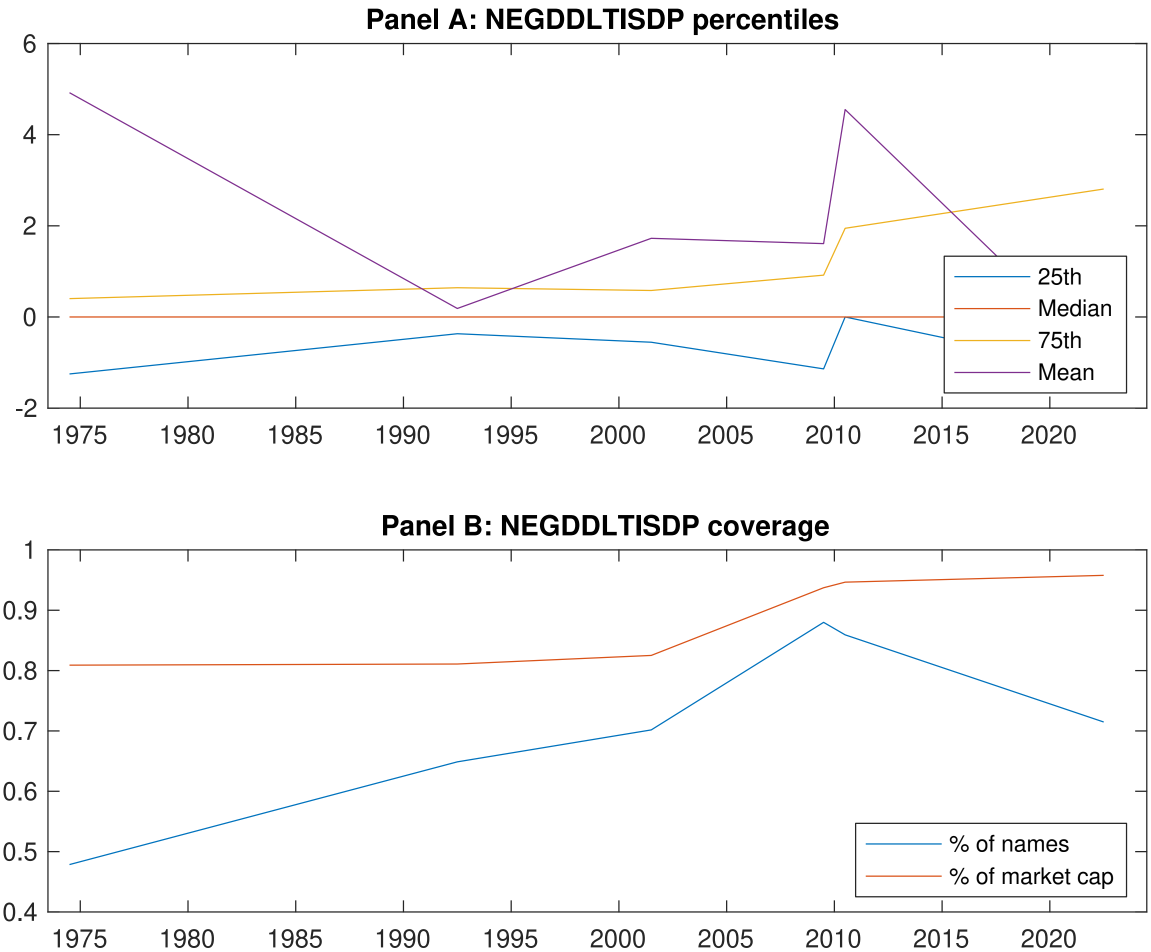



**Figure 1:** Times series of DFP percentiles and coverage.
This figure plots descriptive statistics for DFP. Panel A shows cross-sectional percentiles of DFP over the sample. Panel B plots the monthly coverage of DFP relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DFP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Excess returns and alphas on DFP-sorted portfolios | | | | | | |
|---|---|---|---|---|---|---|
| | (L) | (2) | (3) | (4) | (H) | (H-L) |
| $r^e$ | 0.54 | 0.66 | 0.65 | 0.74 | 0.83 | 0.28 |
| | [2.53] | [3.71] | [3.30] | [4.20] | [4.18] | [3.99] |
| $\alpha_{CAPM}$ | -0.19 | 0.06 | -0.01 | 0.15 | 0.15 | 0.34 |
| | [-3.39] | [1.15] | [-0.19] | [2.87] | [2.99] | [4.92] |
| $\alpha_{FF3}$ | -0.22 | 0.03 | 0.06 | 0.14 | 0.13 | 0.35 |
| | [-4.04] | [0.59] | [1.02] | [2.77] | [2.53] | [4.97] |
| $\alpha_{FF4}$ | -0.17 | 0.03 | 0.10 | 0.10 | 0.12 | 0.29 |
| | [-3.15] | [0.70] | [1.74] | [2.02] | [2.28] | [4.12] |
| $\alpha_{FF5}$ | -0.18 | -0.05 | 0.11 | 0.03 | 0.08 | 0.26 |
| | [-3.36] | [-1.15] | [1.97] | [0.58] | [1.62] | [3.81] |
| $\alpha_{FF6}$ | -0.15 | -0.04 | 0.14 | 0.01 | 0.08 | 0.23 |
| | [-2.77] | [-0.92] | [2.45] | [0.18] | [1.52] | [3.27] |
| Panel B: Fama and French (2018) 6-factor model loadings for DFP-sorted portfolios | | | | | | |
| $\beta_{\text{MKT}}$ | 1.09 | 0.97 | 0.97 | 0.97 | 1.03 | -0.06 |
| | [87.35] | [93.47] | [73.58] | [85.98] | [87.06] | [-3.64] |
| $\beta_{\text{SMB}}$ | 0.08 | -0.11 | -0.03 | -0.02 | 0.14 | 0.06 |
| | [4.46] | [-7.21] | [-1.32] | [-0.96] | [7.81] | [2.30] |
| $\beta_{\text{HML}}$ | 0.14 | 0.08 | -0.15 | -0.09 | -0.00 | -0.14 |
| | [6.02] | [3.83] | [-5.88] | [-4.22] | [-0.03] | [-4.70] |
| $\beta_{\text{RMW}}$ | 0.06 | 0.15 | -0.05 | 0.11 | 0.06 | 0.01 |
| | [2.33] | [7.48] | [-2.12] | [5.10] | [2.71] | [0.19] |
| $\beta_{\text{CMA}}$ | -0.22 | 0.09 | -0.12 | 0.28 | 0.10 | 0.32 |
| | [-6.09] | [3.04] | [-3.07] | [8.62] | [2.81] | [6.82] |
| $\beta_{\text{UMD}}$ | -0.05 | -0.02 | -0.04 | 0.03 | 0.01 | 0.06 |
| | [-4.16] | [-1.48] | [-3.35] | [2.69] | [0.60] | [3.68] |
| Panel C: Average number of firms ($n$) and market capitalization ($me$) | | | | | | |
| $n$ | 667 | 520 | 1039 | 571 | 640 | |
| me ($\$10^6$) | 1646 | 2867 | 2196 | 2886 | 1591 | |

**Table 2:** Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the DFP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Gross Returns and Alphas | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Portfolios | Breaks | Weights | $r^e$ | $\alpha_{\text{CAPM}}$ | $\alpha_{\text{FF3}}$ | $\alpha_{\text{FF4}}$ | $\alpha_{\text{FF5}}$ | $\alpha_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.28 | 0.34 | 0.35 | 0.29 | 0.26 | 0.23 |
| | | | [3.99] | [4.92] | [4.97] | [4.12] | [3.81] | [3.27] |
| Quintile | NYSE | EW | 0.18 | 0.21 | 0.19 | 0.18 | 0.17 | 0.16 |
| | | | [4.04] | [4.75] | [4.45] | [4.07] | [3.89] | [3.70] |
| Quintile | Name | VW | 0.26 | 0.31 | 0.32 | 0.28 | 0.26 | 0.24 |
| | | | [3.71] | [4.44] | [4.58] | [4.01] | [3.79] | [3.48] |
| Quintile | Cap | VW | 0.27 | 0.31 | 0.33 | 0.28 | 0.24 | 0.21 |
| | | | [3.89] | [4.58] | [4.78] | [4.02] | [3.47] | [3.02] |
| Decile | NYSE | VW | 0.24 | 0.30 | 0.29 | 0.20 | 0.15 | 0.10 |
| | | | [2.47] | [3.10] | [2.91] | [2.02] | [1.57] | [1.00] |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas | | | | | | | | |
| Portfolios | Breaks | Weights | $r^e_{net}$ | $\alpha^*_{\text{CAPM}}$ | $\alpha^*_{\text{FF3}}$ | $\alpha^*_{\text{FF4}}$ | $\alpha^*_{\text{FF5}}$ | $\alpha^*_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.22 | 0.28 | 0.28 | 0.25 | 0.21 | 0.18 |
| | | | [3.13] | [3.94] | [3.98] | [3.54] | [3.00] | [2.67] |
| Quintile | NYSE | EW | -0.08 | | | | | |
| | | | [-1.41] | | | | | |
| Quintile | Name | VW | 0.20 | 0.24 | 0.25 | 0.23 | 0.20 | 0.18 |
| | | | [2.84] | [3.47] | [3.58] | [3.29] | [2.93] | [2.71] |
| Quintile | Cap | VW | 0.22 | 0.26 | 0.27 | 0.25 | 0.20 | 0.18 |
| | | | [3.13] | [3.85] | [4.00] | [3.61] | [2.92] | [2.64] |
| Decile | NYSE | VW | 0.17 | 0.23 | 0.22 | 0.17 | 0.11 | 0.07 |
| | | | [1.74] | [2.34] | [2.19] | [1.72] | [1.13] | [0.75] |

**Table 3:** Conditional sort on size and DFP

This table presents results for conditional double sorts on size and DFP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DFP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DFP and short stocks with low DFP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

| | | DFP Quintiles | | | | | DFP Strategies | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (L) | (2) | (3) | (4) | (H) | $r^e$ | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
| Size quintiles | (1) | 0.66 | 0.89 | 0.97 | 0.96 | 0.75 | 0.10 | 0.12 | 0.09 | 0.06 | 0.05 | 0.03 |
| | | [2.41] | [3.23] | [3.53] | [3.44] | [2.67] | [1.04] | [1.26] | [0.94] | [0.65] | [0.48] | [0.34] |
| | (2) | 0.73 | 0.90 | 0.83 | 0.93 | 0.84 | 0.11 | 0.15 | 0.12 | 0.12 | 0.08 | 0.08 |
| | | [2.73] | [3.58] | [3.28] | [3.76] | [3.35] | [1.50] | [1.99] | [1.65] | [1.52] | [1.04] | [1.02] |
| | (3) | 0.82 | 0.76 | 0.84 | 0.82 | 0.91 | 0.10 | 0.15 | 0.15 | 0.11 | 0.14 | 0.12 |
| | | [3.26] | [3.43] | [3.51] | [3.62] | [3.89] | [1.23] | [1.89] | [1.85] | [1.40] | [1.78] | [1.47] |
| | (4) | 0.71 | 0.81 | 0.83 | 0.76 | 0.87 | 0.16 | 0.19 | 0.18 | 0.17 | 0.13 | 0.13 |
| | | [3.17] | [3.84] | [3.73] | [3.58] | [4.06] | [2.13] | [2.61] | [2.45] | [2.28] | [1.75] | [1.69] |
| | (5) | 0.45 | 0.62 | 0.60 | 0.67 | 0.82 | 0.37 | 0.40 | 0.41 | 0.34 | 0.32 | 0.27 |
| | | [2.24] | [3.45] | [3.16] | [3.67] | [4.25] | [4.10] | [4.42] | [4.67] | [3.78] | [3.64] | [3.06] |

Panel B: Portfolio average number of firms and market capitalization

| | | DFP Quintiles | | | | | DFP Quintiles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Average $n$ | | | | | Average market capitalization ($\$10^6$) | | | | |
| | | (L) | (2) | (3) | (4) | (H) | (L) | (2) | (3) | (4) | (H) |
| Size quintiles | (1) | 386 | 386 | 388 | 386 | 383 | 36 | 33 | 32 | 32 | 35 |
| | (2) | 106 | 106 | 105 | 106 | 105 | 59 | 61 | 59 | 60 | 60 |
| | (3) | 75 | 75 | 75 | 75 | 75 | 107 | 107 | 103 | 105 | 106 |
| | (4) | 63 | 63 | 63 | 63 | 63 | 227 | 235 | 226 | 232 | 225 |
| | (5) | 58 | 58 | 58 | 58 | 58 | 1448 | 2040 | 1875 | 2220 | 1464 |

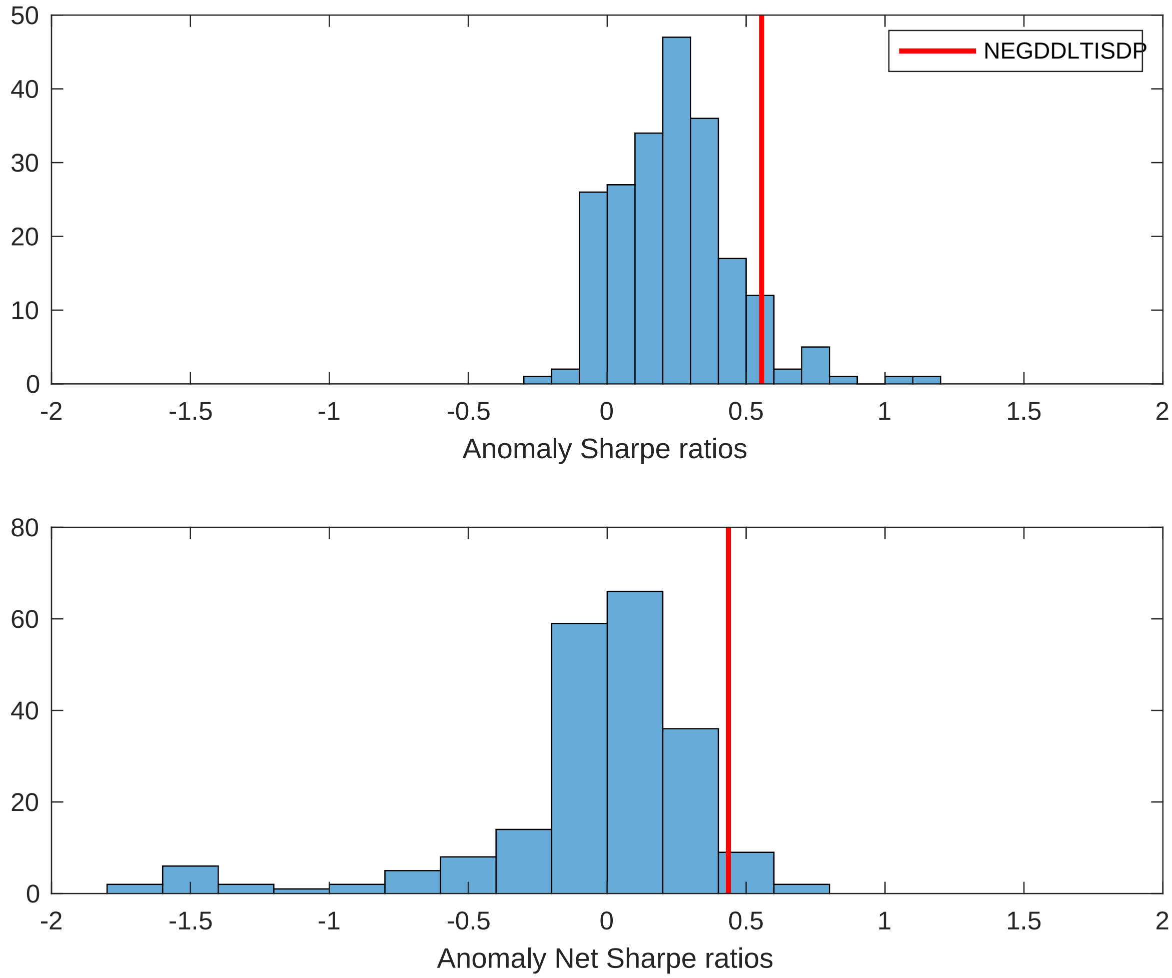



**Figure 2:** Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DFP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

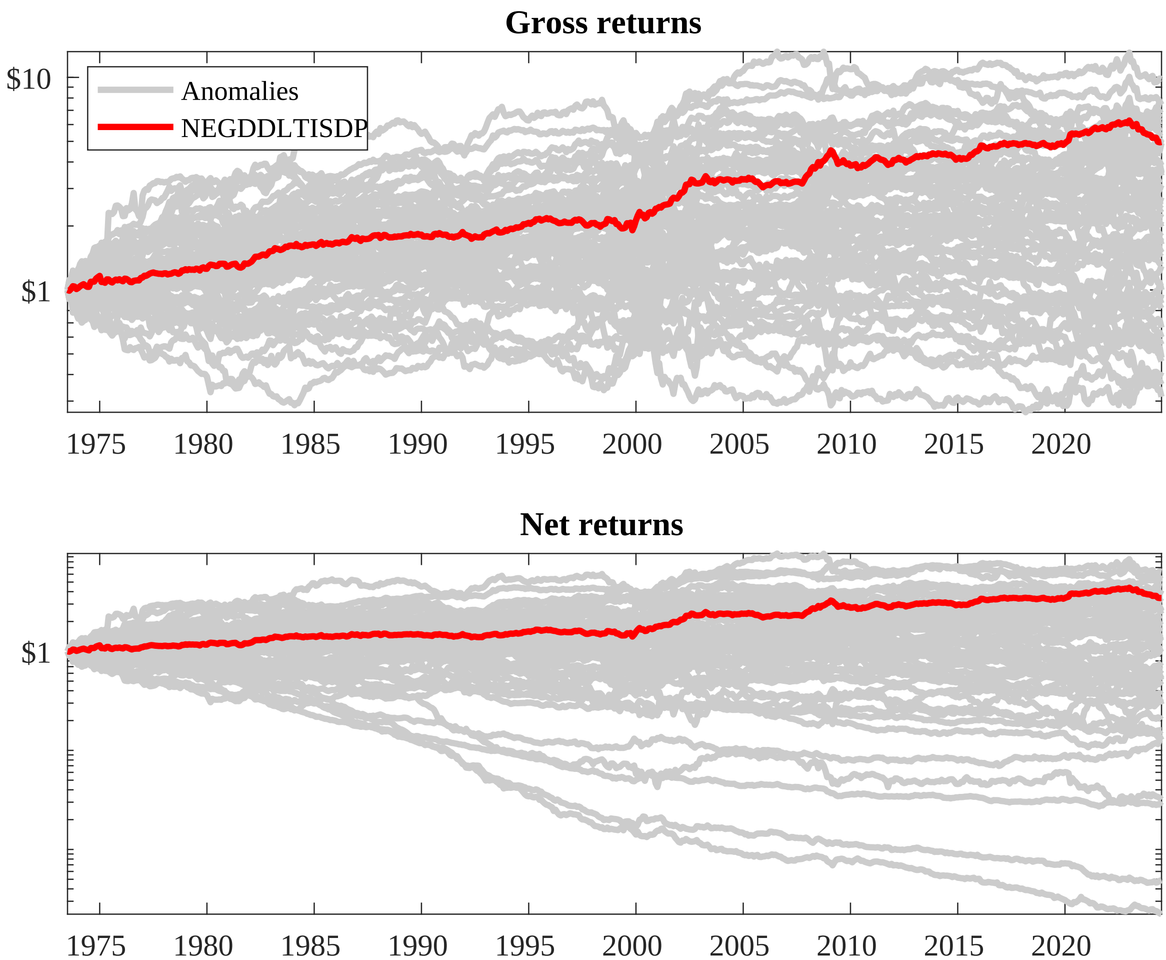
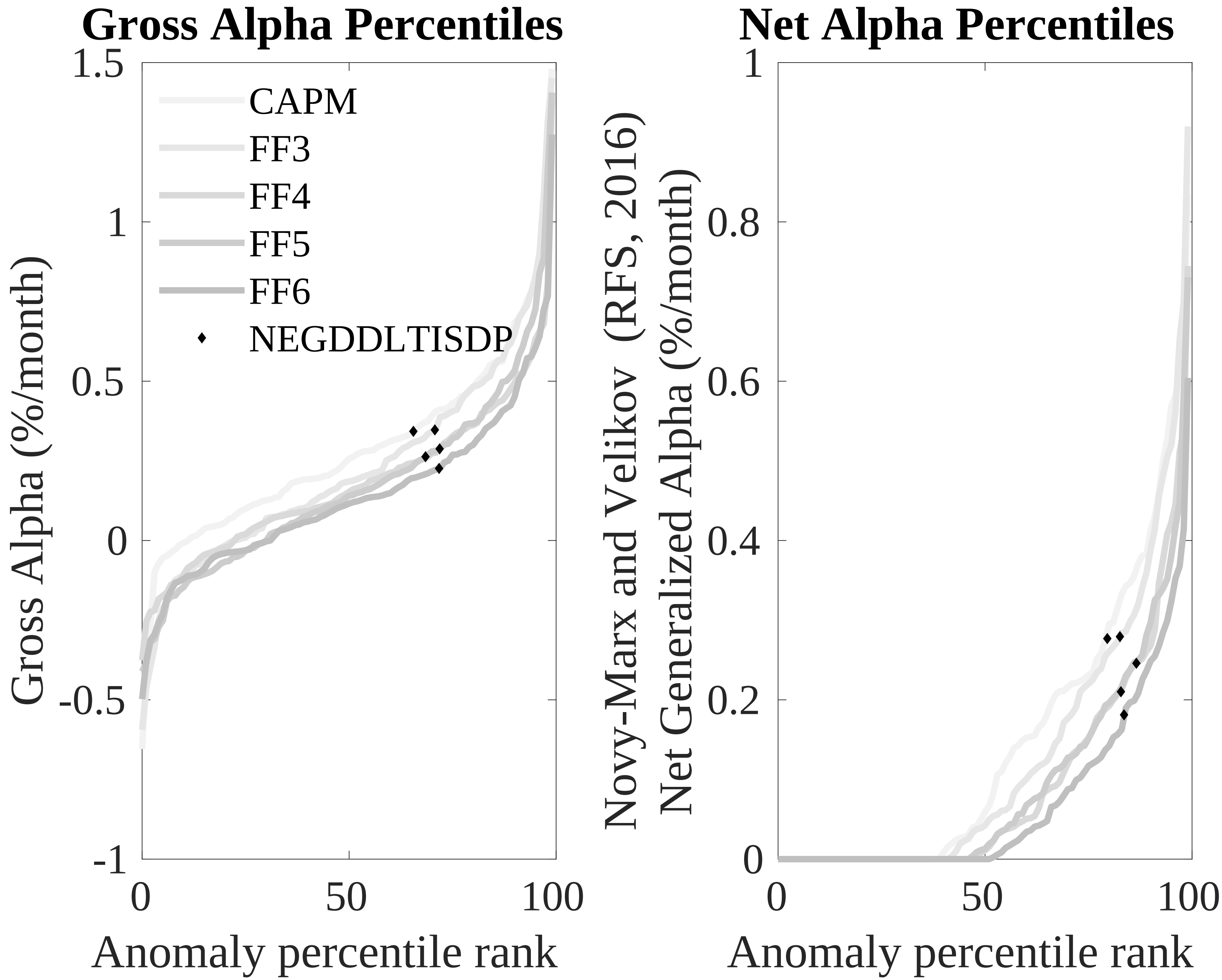
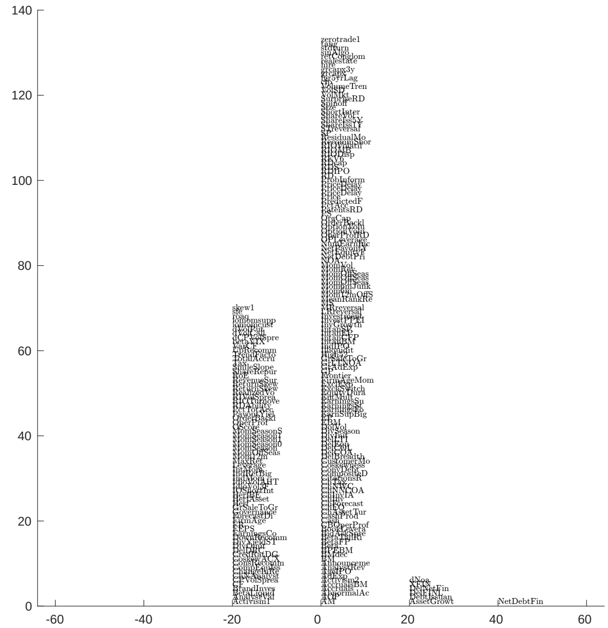
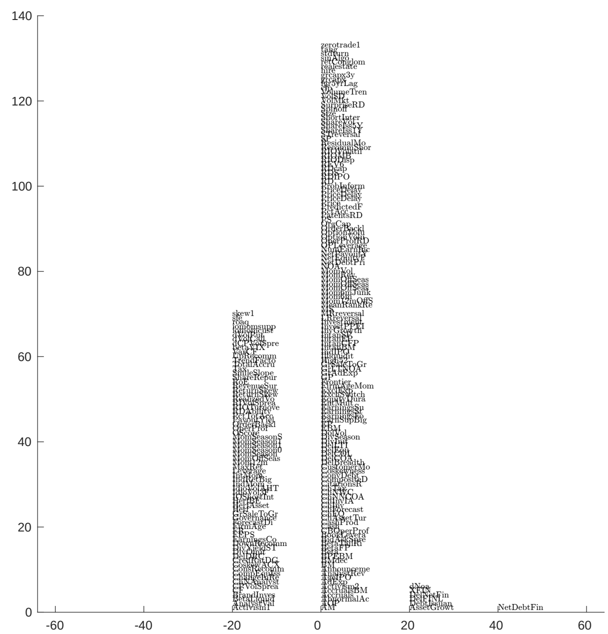



**Figure 3:** Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DFP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISDP
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



Panel A: Pearson



Panel B: Spearman

**Figure 5:** Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with DFP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

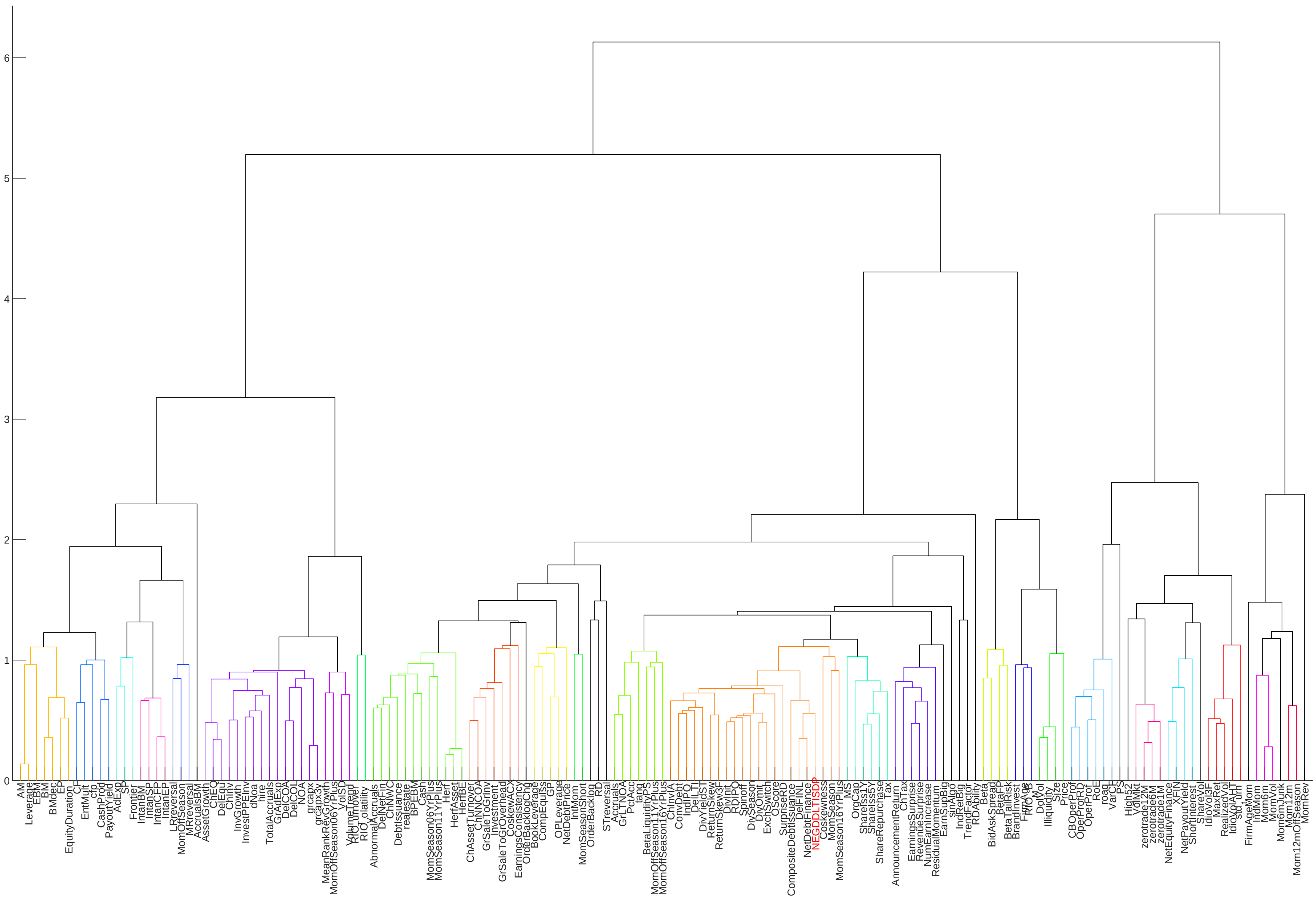



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

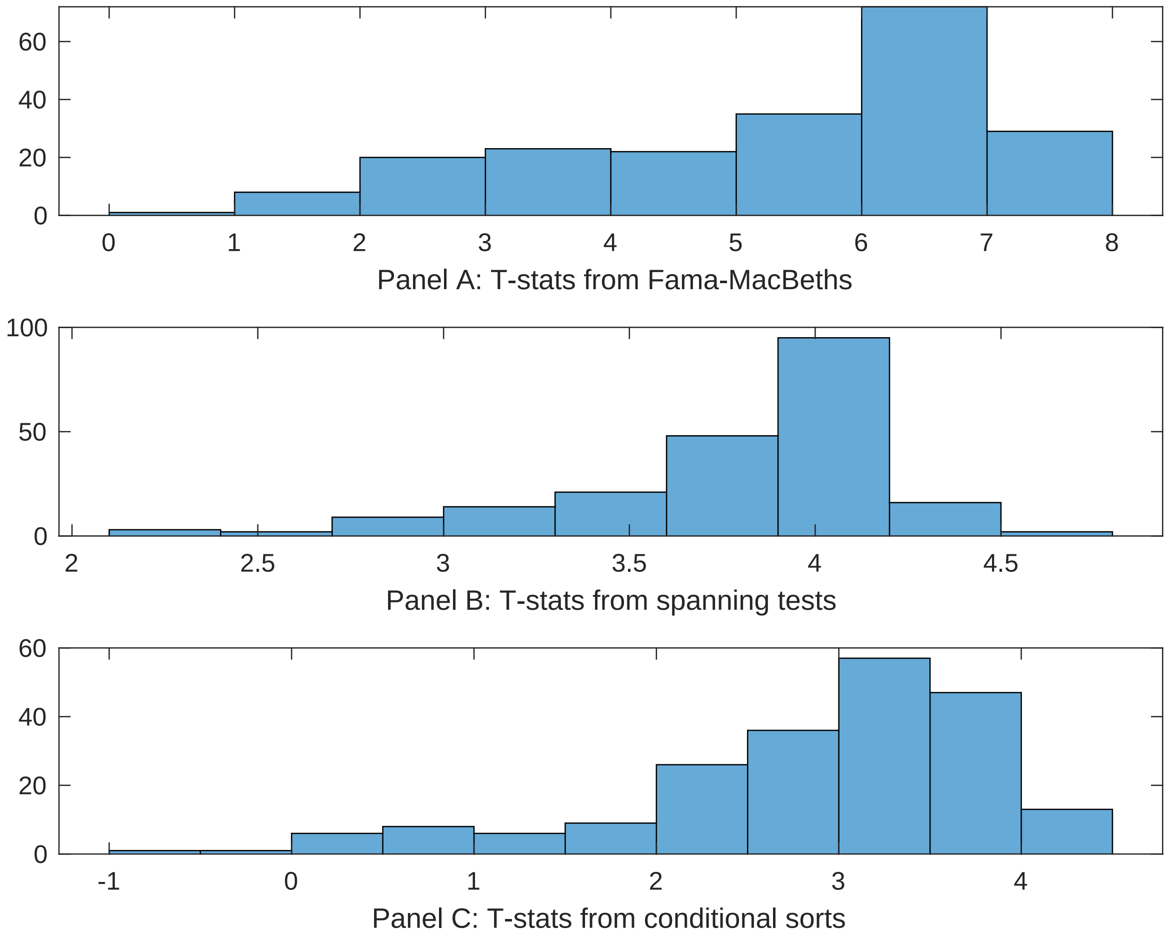



**Figure 7:** Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of DFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{DFP}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DFP} DFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{DFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DFP. Stocks are finally grouped into five DFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DFP trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on DFP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DFP} DFP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, $X$, are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.13 | 0.13 | 0.14 | 0.14 | 0.14 | 0.13 | 0.15 |
| | [5.39] | [5.36] | [5.72] | [5.82] | [5.31] | [5.37] | [5.65] |
| DFP | 0.46 | 0.58 | 0.76 | 0.38 | 0.12 | 0.11 | 0.16 |
| | [2.32] | [2.70] | [3.65] | [1.79] | [5.37] | [5.59] | [0.62] |
| Anomaly 1 | 0.17 | | | | | | -0.58 |
| | [9.28] | | | | | | [-1.26] |
| Anomaly 2 | | 0.20 | | | | | 0.93 |
| | | [8.70] | | | | | [1.38] |
| Anomaly 3 | | | 0.19 | | | | 0.10 |
| | | | [6.23] | | | | [1.74] |
| Anomaly 4 | | | | 0.11 | | | 0.73 |
| | | | | [8.47] | | | [3.52] |
| Anomaly 5 | | | | | 0.40 | | 0.93 |
| | | | | | [6.88] | | [1.68] |
| Anomaly 6 | | | | | | 0.92 | 0.92 |
| | | | | | | [6.05] | [0.66] |
| # months | 612 | 612 | 612 | 612 | 612 | 612 | 612 |
| $\bar{R}^2(\%)$ | 0 | 0 | 1 | 0 | 0 | 0 | 0 |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DFP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DFP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where $X_k$ indicates each of the six most-closely related anomalies and $f_j$ indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, $X$, are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.19 | 0.20 | 0.20 | 0.21 | 0.22 | 0.22 | 0.18 |
| | [2.88] | [2.95] | [2.93] | [3.12] | [3.14] | [3.10] | [2.66] |
| Anomaly 1 | 18.20 | | | | | | 9.43 |
| | [4.74] | | | | | | [1.85] |
| Anomaly 2 | | 20.55 | | | | | 12.21 |
| | | [5.58] | | | | | [2.40] |
| Anomaly 3 | | | 11.88 | | | | 8.87 |
| | | | [3.47] | | | | [2.42] |
| Anomaly 4 | | | | 0.78 | | | -3.74 |
| | | | | [0.19] | | | [-0.88] |
| Anomaly 5 | | | | | 1.83 | | 1.44 |
| | | | | | [0.68] | | [0.52] |
| Anomaly 6 | | | | | | 0.28 | -1.51 |
| | | | | | | [0.08] | [-0.39] |
| mkt | -5.11 | -5.49 | -3.91 | -5.55 | -5.58 | -5.54 | -4.14 |
| | [-3.27] | [-3.55] | [-2.38] | [-3.49] | [-3.51] | [-3.48] | [-2.52] |
| smb | 3.69 | 3.74 | 9.37 | 5.51 | 5.76 | 5.58 | 6.68 |
| | [1.54] | [1.58] | [3.58] | [2.28] | [2.38] | [2.31] | [2.41] |
| hml | -13.28 | -13.65 | -13.16 | -14.49 | -14.52 | -14.51 | -11.97 |
| | [-4.40] | [-4.57] | [-4.31] | [-4.72] | [-4.74] | [-4.61] | [-3.85] |
| rmw | -0.86 | -1.48 | -6.32 | 0.78 | 1.05 | 0.79 | -6.59 |
| | [-0.28] | [-0.48] | [-1.71] | [0.25] | [0.33] | [0.25] | [-1.77] |
| cma | 26.40 | 27.17 | 24.74 | 31.52 | 30.84 | 32.21 | 25.00 |
| | [5.57] | [5.86] | [4.84] | [4.70] | [5.91] | [5.65] | [3.56] |
| umd | 4.42 | 4.67 | 5.91 | 6.02 | 5.81 | 5.98 | 4.20 |
| | [2.79] | [3.00] | [3.78] | [3.81] | [3.63] | [3.75] | [2.61] |
| # months | 612 | 612 | 612 | 612 | 612 | 612 | 612 |
| $\bar{R}^2(\%)$ | 19 | 20 | 17 | 16 | 16 | 16 | 20 |

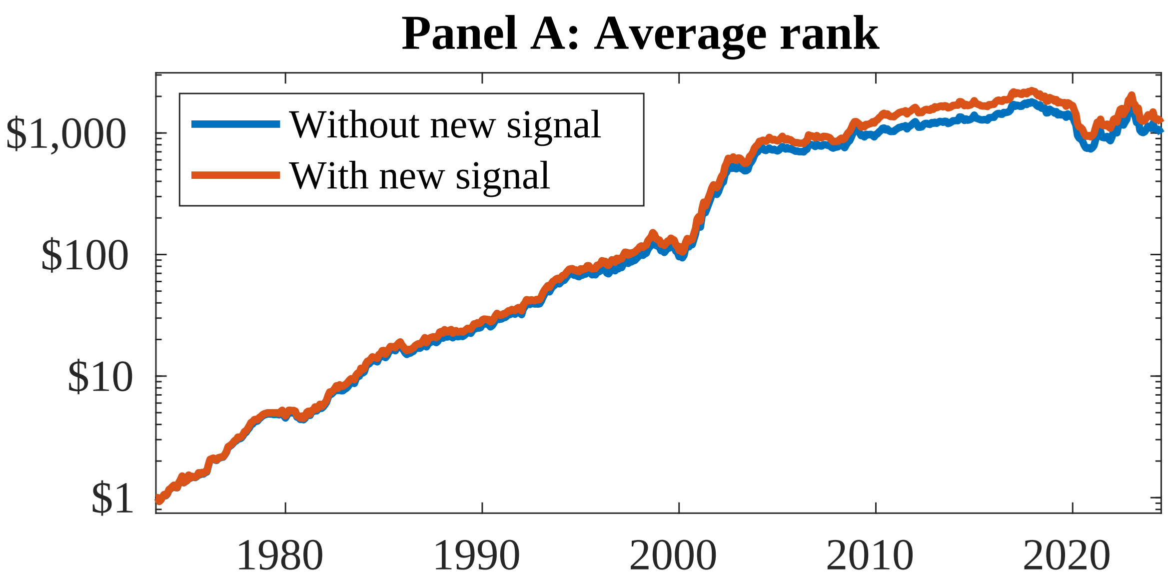



**Figure 8:** Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DFP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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